Cambridge International AS & A Level

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CHEMISTRY **9701/52**

Paper 5 Planning, Analysis and Evaluation **October/November 2017**

MARK SCHEME

Maximum Mark: 30

Published

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Question	Answer	Marks
1(a)(i)	$CuCO_3$ and $Cu(OH)_2$ both react (with HC*l*) or both form copper(II) chloride	**1**
1(a)(ii)	(Transfer) 12.5(0) cm^3 of (10.0 mol dm^{-3}) HC*l* using a (graduated) pipette or a burette	**1**
	add to a 250 cm^3 volumetric flask **AND** make to mark with distilled water	**1**
1(a)(iii)	Measure a volume of gas from the carbonate reaction **or** measure the (loss of) mass from the carbonate reaction	**1**
1(a)(iv)	Suitable apparatus for production of CO_2	**1**
	Suitable means of measuring CO_2 evolved	**1**
1(a)(v)	Correct labels on axes *y*-axis: volume (of gas) or mass loss or mass of 'limewater' **and** *x*-axis: time or t	**1**
	curved line (from origin) to reach a plateau, e.g. 	**1**

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Question	Answer	Marks
1(a)(vi)	Any sensible attempt seen to make the experiment accurate If mass loss Reduce risk of mass loss through spraying Insert cotton wool plug If gas collection Any method to reduce risk of gas loss Check apparatus is sealed Insert bung quickly Any attempt to measure temperature Check apparatus is at room temperature Apparatus accuracy Use an accurate or 2dp (or more) balance / gas syringe / measuring cylinder	**1**
1(a)(vii)	mol of $CuCO_3 = 0.5 \div 123.5 = 4.05 \times 10^{-3}$ mol	**1**
	moles of HC$l = 2 \times 4.05 \times 10^{-3} = 8.10 \times 10^{-3}$ mol **and** volume of HC$l = 8.10 \times 10^{-3} \div 0.500 = 0.0162\ dm^3$ $= 16.2\ cm^3$	**1**

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Question	Answer	Marks
1(b)	Any suitable precaution relating to stated hazard of given chemical For HCl Precaution (lab) gloves Explanation (10 mol dm^{-3}) HCl is corrosive For $CuCO_3$ Precaution (lab) gloves / wash hands (after use) / face or mouth mask Explanation Harmful if swallowed	**1**
1(c)(i)	moles of $H_2SO_4 = 0.40 \times \frac{24.15}{1000} = 9.66 \times 10^{-3}$ mol	**1**
	mass of $Cu_3(CO_3)_2(OH)_2 = 344.5 \times 9.66 \times 10^{-3} \div 3 = 1.11$ g	**1**
	% by mass = $\frac{1.11}{1.50} \times 100\% = 74.0\%$	**1**

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Question	Answer	Marks
1(c)(ii)	Problem 1 titres are not concordant / are too far apart / are 0.5(0) **cm^3** apart / difference is too large Improvement Repeat until (two) concordant titres have been achieved / two readings within 0.1(0) **cm^3** Problem 2 colour change (of indicator) will be masked Improvement 2 Use an alternative indicator / named indicator [1] for each problem, [1] for an improvement	**3**

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<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>2(a)(i)</td><td>
<table>
<tr><th>Difference in conc. D</th><th>$\frac{D}{m}$</th><th>$\log(\frac{D}{m})$</th><th>log[X]</th></tr>
<tr><td>24.04</td><td>120.20</td><td>2.08</td><td>–0.02</td></tr>
<tr><td>24.31</td><td>97.24</td><td>1.99</td><td>–0.16</td></tr>
<tr><td>24.40</td><td>81.33</td><td>1.91</td><td>–0.22</td></tr>
<tr><td>24.59</td><td>70.26</td><td>1.85</td><td>–0.39</td></tr>
<tr><td>24.67</td><td>61.68</td><td>1.79</td><td>–0.48</td></tr>
<tr><td>24.73</td><td>54.96</td><td>1.74</td><td>–0.57</td></tr>
<tr><td>24.77</td><td>49.54</td><td>1.69</td><td>–0.64</td></tr>
<tr><td>24.80</td><td>45.09</td><td>1.65</td><td>–0.70</td></tr>
<tr><td>24.83</td><td>41.38</td><td>1.62</td><td>–0.77</td></tr>
</table>
D data correct [1]
log[X] data correct [1]
All data to 2 dp [1]
</td><td>3</td></tr>
<tr><td rowspan="2">2(a)(ii)</td><td>greater adsorption</td><td>1</td></tr>
<tr><td>greater surface area available</td><td>1</td></tr>
<tr><td rowspan="2">2(b)</td><td>all nine points plotted correctly</td><td>1</td></tr>
<tr><td>best-fit straight line drawn</td><td>1</td></tr>
<tr><td rowspan="2">2(c)</td><td>Correct point (at –0.22, 1.91) identified</td><td>1</td></tr>
<tr><td>Statement explaining lack of adsorption, e.g.
not enough stirring,
mass of activated charcoal too low,
surface area not high enough / too low / coagulation of charcoal / bulkier particles used
not left long enough</td><td>1</td></tr>
</table>

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Question	Answer	Marks
2(d)(i)	co-ordinates read and recorded correctly	**1**
	gradient determined **and** same value for *b*	**1**
2(d)(ii)	intercept on *y*-axis read and recorded correctly	**1**

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